JATIN CHOPRA

Personal Information

Full Name: Jatin Chopra **Nationality:** Indian **Degree Programme:** MSc Astrophysics

Current Institution: Rheinische Friedrich-Wilhelms-Universität Bonn
(Expected) Graduation Date: 20th November 2026

Contact: My Webpage | My GitHub Page | LinkedIn | jchopra@astro.uni-bonn.de | choprajatin4421@gmail.com

Education

Rheinische Friedrich-Wilhelms-Universität Bonn Bonn, Germany
MSc Astrophysics October 2023 – November 2026

Savitribai Phule Pune University Pune, India
BSc Physics (Minor in Mathematics and Electronics) July 2019 – September 2022
CGPA: 9.77/10.0

Research Internships

1. **Estimating Initial Cloud Conditions from Molecular Abundances using Neural Networks and Bayesian Inference** *October 2025 – Ongoing*
   - Supervised by Prof. Serena Viti and Dr. Gijs Vermariën at the Argelander-Institut für Astronomie (AIfA) and Universiteit Leiden. Developing a neural network in **JAX** to emulate UCLCHEM.
   - The emulator is used as a forward model in a **NumPyro**-based Bayesian inference routine (SA sampler with 4 chains) on real gas and ice observations, to estimate initial cloud conditions, including temperature, final density, cosmic ray ionisation rate, and UV background radiation.
   - Investigating the effect of incorporating survival function, CDF, and PDF for molecules with only upper-limit abundance measurements, improving uncertainty estimation of molecules with weak observations.
2. **Studying the Role of Galaxy Spheroids in Star Formation** *May 2025 – July 2025*
   - Supervised by Prof. Frank Bigiel and Dr. Dario Colombo at the Argelander-Institut für Astronomie (AIfA).
   - Built a Python pipeline for multicomponent photometric decomposition (Sérsic, bulge+disk, bulge+bar+disk, bulge+broken disk, bulge+bar+broken disk) of 643 galaxies (**iEDGE** survey), extracting Bulge-to-Total (B/T) light ratios, using **photutils**, **sep**, and **SciPy** (least-squares model fitting) to study the influence of morphology on star formation quenching.
   - Currently working on this as a side project to estimate the bulge photometric mass (using curve_fit and **emcee**).
3. **Scattering of Methanol Masers and Electron Distribution of the Milky Way** *December 2024 – May 2025*
   - Research intern under Prof. Amélie Saintonge and Dr. Andreas Brunthaler at the Max Planck Institute for Radio Astronomy, Bonn. Reduced and analysed **VLBA** data using **AIPS** and **ParselTongue** to measure angular broadening of 21 CH_3OH masers (taken from **BeSSeL**) and 10 background quasars.
   - Compared measured scattering against the **NE2001** electron density model via dispersion measure calculations. Observed discrepancies between the model predictions and the measured scattering for masers, which were attributed to intrinsic maser sizes. No discrepancies were observed for quasars.
4. **Detecting and Classifying AGNs in MaNGA Galaxies using Resolved BPT Diagnostics** *July 2024 - September 2024*
   - Developed a **Python+SQL** pipeline to detect AGNs from the **MaNGA** survey by making resolved BPT maps and classifying AGNs based on Hα equivalent width into strong/weak.

* Detected 567 AGNs from 10,070 MaNGA galaxies, with good classification overlap with broad, X-ray, and IR AGN catalogues; this project was supervised by Prof. Frank Bigiel and Dr. Mallory Thorp at the Argelander-Institut für Astronomie (AIfA).

Master's Thesis

Title: Star-Formation Quenching in MaNGA Post-Merger Galaxies

Supervisors: Prof. Dr. Frank Bigiel and Dr. Mallory Thorp
Examiners: Prof. Dr. Frank Bigiel and Prof. Dr. Amélie Saintonge

Submission: 20th November 2026

Thesis Abstract

Simulations and observational studies have shown that galaxy mergers play a crucial role in galaxy evolution. Recent spatially resolved studies show that merger-triggered starbursts exhibit distinct radial star formation profiles compared to their secular counterparts. Observationally, post-mergers include a large number of galaxies which recently stopped forming stars. The connection between mergers and the processes through which galaxies stop forming stars (called star formation quenching) is less constrained. To discern whether mergers present a unique route to quenching, this thesis investigated whether mergers leave a unique signature in kiloparsec (kpc)-scale star formation profiles.

Integral field spectroscopy (IFS) data from the Mapping Nearby Galaxies at Apache Point Observatory (MaNGA) Survey (10,070 Galaxies) were used to study the extent of quenching in a galaxy and classify it into quenching stages. Quenching stages are unique kpc-scale signatures in the ionised gas of a galaxy (following Kalinova et al. (2021)). All MaNGA galaxies were classified into their respective quenching stages using an automated and visual classification method. Accurate star formation maps and quenching classifications were determined for 388 post-mergers, which were compared to 5928 isolated galaxies. The post-mergers were adopted from Thorp et al. (2024), where they were visually identified.

The distributions of post-mergers and isolated galaxies are broadly similar across the quenching stages, except that post-mergers have an excess in the star-forming case and a deficiency in fully retired galaxies. Thus suggesting that mergers enhance star formation and delay complete retirement. The kpc-scale star formation profiles of the two samples were investigated by matching each post-merger to at least five isolated control galaxies with similar properties at the same quenching stage. Radial profiles of $\Delta\Sigma_{\mathrm{SFR}}$ (Offset from the resolved star formation main sequence, calculated using Hα emission + Dn4000 index) were created by comparing spaxels of post-mergers and controls that occupy similar regions within a galaxy. Within each quenching stage, the shape of $\Delta\Sigma_{\mathrm{SFR}}$ profiles generally agrees between post-mergers and controls, except for certain unique regions in post-mergers that show more efficient quenching.

The similarities between post-merger and control radial profiles imply that the process of quenching is relatively consistent except in regions of enhanced quenching. This is unlike starburst behaviour, which leaves a uniquely strong signature in post-merger centres. A spatial coexistence of active galactic nuclei (AGNs) and quenched regions was observed for both post-mergers and controls, suggesting that AGN activity alone does not drive the quenching in mergers. The influence of morphology was investigated using the Sérsic index. The dominance of morphological quenching can be ruled out, as the Sérsic indices of post-mergers and controls agree within uncertainties at each quenching stage. Overall it is concluded that although mergers are associated with boosted star formation and delayed retirement, the spatial progression of quenching largely follows patterns observed in isolated galaxies, except for some regions of efficient quenching.

Teaching Experience

1. **Argelander-Institut für Astronomie (AIfA)** August 2025 - November 2026
   Wissenschaftliche Hilfskraft (Tutor of Optical Astronomy Exoplanet Transit Detection Lab), Bonn, Germany

   * Explaining the observation, calibration, and data reduction of a modern optical telescope, conducting oral examinations, carrying out telescope observation, supervising data reduction, and grading lab reports of MSc students. Additionally, giving tours of the telescope on the rooftop of AIfA to visiting students and guests.

   * Contributing to the astrophotography done at AIfA by observing Milky Way objects and other deep-sky objects.

2. **Naxxatra Sciences and Collaborative Research** January 2023 - February 2023
   Research and Teaching Fellow, Bangalore, India

   * Conducted hands-on astrophysics lectures for 300 students (grades 5-12), on topics like galaxy morphology, the cosmic distance ladder, the working of different telescopes, methods to detect exoplanets, and introduced dark matter and CMB.

   * Performed experiments like sunspot tracking, acceleration due to gravity measurement, and Earth's circumference calculation. Supervised Solar observations. Conducted stargazing sessions and comet C/2022 E3 (ZTF) observation nights for students, faculty, and parents.

Scientific Training

1. **Attended the 2nd European ALMA Winter School 2026** January 2026
   The school was organised by the University of Leiden, which covered the calibration, imaging, proposal writing, and reduction pipeline to study the astronomical data observed with the ALMA telescope. I was one of the few master's students to attend this school. I gained an improved understanding of ALMA data reduction using **CASA**, **GILDAS**, and got familiarised with imaging beyond the traditional CLEAN method.

2. **Attended the Physics of the Early Universe 2022** January 2022
   I was one of the very few undergraduate students selected to attend this program organised by the International Centre for Theoretical Sciences (ICTS) Bangalore. Studied advanced topics such as Perturbation Theory, Dark Energy, Inflation, CMB anisotropies, Galaxy Formation, and Big Bang Nucleosynthesis.

Conferences and Post-Presentation

1. **Astronomische Gesellschaft (AG) 2026** September 2026
   Gave a splinter talk and presented a poster on my master's thesis work at the AG 2026 annual meeting. View abstract

2. **BCGS 20th Anniversary Celebration** November 2026
   I will present a poster on my master's thesis work at the Bonn-Cologne Graduate School of Physics and Astronomy (BCGS) 20th Anniversary Celebration meeting.

Honours and Awards

1. **Highest Grade in the College Department, BSc Physics** March 2023
   Graduated with the top grade in my department at N.W. College of Savitribai Phule Pune University and was awarded a departmental scholarship in recognition of this performance.

2. **Discovered Two Asteroids** September 2021
   I detected near-Earth objects and main-belt asteroids in data from the Pan-STARRS survey, provided by the International Astronomical Search Collaboration (IASC). Made 12 preliminary discoveries and 2 provisional discoveries, including two main-belt asteroids, namely, **2021 RZ8** and **2021 QO24**.

Skills

* **Programming Languages:** Python, SQL, C++, HTML, Fortran
* **Python Libraries:** Astropy, SciPy, pymysql, ParselTongue, JAX, Equinox, Optax, NumPyro, PyTorch, Scikit-learn, sep, photutils, astroquery, NumPy, Pandas, Matplotlib, Seaborn
* **Astrophysics Software:** CASA, AIPS, MESA, DS9, GILDAS, PHOEBE, Period04, Astrometrica, NINA
* **Surveys & Telescopes:** MaNGA, BeSSeL, iEDGE, SDSS, Pan-STARRS, VLBA, ALMA, UCLCHEM (chemical model)
* **Other Computational Skills:** Linux, LaTeX, GitHub, Microsoft Office
* **Soft Skills:** Pedagogy and Science Communication, Leadership and Team Collaboration, Analytical Problem Solving, Rapid Skill Acquisition, Resilience and Discipline, Detail-Oriented and Self-Motivated
* **Languages:** Hindi (Native Speaker), English, German, Punjabi, and Marathi.

Other Relevant Experience

1. **Astrophotography at Universität Bonn** *August 2025 - Ongoing*
   Website: Astrophotography Jatin Chopra
   Conducted frequent deep-sky observations using the 35 cm optical telescope at AIfA. Observed objects like NGC 891, M51, M81, M82, M1, and NGC 7331.
2. **Writer at The Universe 101** *December 2020 - April 2021*
   Website: The Universe 101
   Began an astrophysics blog to explain cosmic mysteries in simple terms. Articles like "Cosmic Alchemy" and "Dark Matter" received decent attention. I plan to restart the blog and eventually transform it into a YouTube channel.

References

* **Prof. Frank Bigiel**
  Argelander-Institut für Astronomie, Rheinische Friedrich-Wilhelms-Universität Bonn
  Professor of Astrophysics | bigiel@astro.uni-bonn.de
* **Prof. Serena Viti**
  Argelander-Institut für Astronomie, Rheinische Friedrich-Wilhelms-Universität Bonn
  Professor of Astrochemistry | sviti@uni-bonn.de
* **Dr. Mallory Thorp**
  Argelander-Institut für Astronomie, Rheinische Friedrich-Wilhelms-Universität Bonn
  Postdoctoral Researcher | mthorp@astro.uni-bonn.de
* **Dr. Kiran Wani**
  Indian Institute of Astrophysics, Bangalore
  Postdoctoral Researcher | kiran.wani@iiap.res.in